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# From Tokens to Policy: Actionable Heterogeneous Treatment Effect Identification

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## Abstract

1 [All: WAIT TO READ, rewrite completely] Modern experiments can now be en-  
2 riched with images, text, and other complex pre-treatment observations alongside  
3 standard covariates. This creates new opportunities for heterogeneous-treatment-  
4 effect characterization, but also a redundancy problem: extracted concepts may  
5 contain many overlapping proxies of the same underlying heterogeneity driver.  
6 We formalize the goal as learning an interpretable treatment-effect representa-  
7 tion, namely one minimal sufficient subset of observed concepts whose treatment-  
8 dependent information captures all heterogeneity recoverable from the full candi-  
9 date concept space. This reveals a mismatch with vanilla coordinate-wise multiple  
10 testing, which instead targets all marginally active concepts and can become in-  
11 creasingly redundant as experimental power grows. We propose *Neural Effect*  
12 *Modifier Search* (NEMS), a recursive conditional-testing procedure for compact  
13 representative recovery, and study it on synthetic benchmarks and on real exper-  
14 imental and quasi-experimental analyses augmented with complex pre-treatment  
15 observations.

Placeholder

Figure 1: Pipeline Illustration

## 1 Introduction

17 [All: WAIT TO READ] Characterizing heterogeneous treatment effects is essential for understand-  
18 ing not only whether an intervention works on average, but also *for whom* it helps or harms [Roth-  
19 man et al., 1980, Gail and Simon, 1985]. This question is already statistically delicate when candi-  
20 date effect modifiers are low-dimensional and interpretable, because one must distinguish genuine  
21 treatment-effect heterogeneity from noise and from purely prognostic variation in outcomes. It  
22 becomes even harder when experiments are enriched with high-dimensional, unstructured, or multi-  
23 modal pre-treatment observations, for example by complementing standard baseline covariates with  
24 satellite imagery, medical images, or text [Jerzak et al., 2023].

25 This setting is only partially addressed by existing heterogeneous-treatment-effect methods. Flexi-  
26 ble approaches such as meta-learners [Künzel et al., 2019], forest methods [Wager and Athey, 2018,

27 Atthey et al., 2019], orthogonal or debiased learners [Nie and Wager, 2021], Bayesian nonparamet-  
 28 rics [Hill, 2011], and representation-learning approaches [Shalit et al., 2017] can estimate complex  
 29 heterogeneity patterns, but they are not primarily designed to identify which pre-treatment concepts  
 30 drive such heterogeneity. Conversely, more interpretable approaches typically assume that candidate  
 31 effect modifiers are already available as structured covariates [Atthey and Imbens, 2016, Bargagli-  
 32 Stoffi et al., 2020]. The aim here is instead to start from a candidate concept space built from mixed  
 33 pre-treatment observations, structured and unstructured, and recover a *non-redundant representative*  
 34 *set* of observed concepts whose treatment-dependent information captures all heterogeneity recov-  
 35 erable from that candidate space.

36 A central difficulty is redundancy. Once candidate concepts are extracted from rich pre-treatment  
 37 observations, several observed coordinates may appear to modify the treatment effect simply be-  
 38 cause they encode overlapping information about the same latent source of heterogeneity. In that  
 39 regime, vanilla coordinate-wise multiple testing becomes increasingly difficult to interpret: as exper-  
 40 imental power grows, more and more coordinates may be detected, even though many of them are  
 41 only redundant proxies. The challenge is therefore not only to detect heterogeneity, but to recover a  
 42 *sufficient* and *minimal* characterization of it.

**Motivating problem: Uganda Youth Opportunities Programme.** Consider an intervention that provided disadvan-  
 tagged young adults with grants to support vocational training and business activities [Blattman et al., 2014]. Beyond ask-  
 ing whether the program works on average, we would like to characterize *which kinds of participants* benefit most. Stan-  
 dard baseline surveys provide interpretable covariates, but other  
 pre-treatment factors, such as remoteness, local infrastructure,  
 or environmental surroundings, may also shape heterogeneity  
 and may be partially observable from satellite imagery [Jerzak  
 et al., 2023]. Once survey-based and image-derived concepts  
 are combined, several observed coordinates may encode over-  
 lapping aspects of the same underlying driver. The challenge is  
 to integrate such multimodal pre-treatment information while  
 still identifying a minimal and sufficient characterization of  
 treatment-effect heterogeneity.

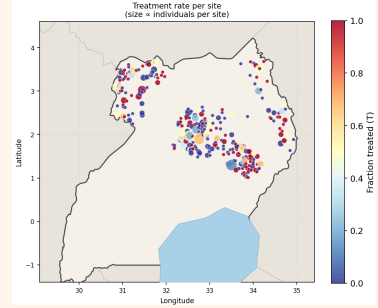


Figure 2: Uganda Youth Opportunities Programme geographic illustration.

43  
 44 We refer to this target as *representative recovery*. The goal is not to return every observed concept  
 45 carrying some heterogeneity signal, but rather one minimal sufficient subset of observed concepts  
 46 that still captures all treatment-effect heterogeneity recoverable from the available candidate space.

47 To address this problem, we propose *Neural Effect Modifier Search* (NEMS), a recursive conditional-  
 48 testing procedure that searches for a compact representative set rather than an exhaustive list  
 49 of marginal discoveries. Practically, NEMS can be combined with modern concept-construction  
 50 pipelines, including representation learning or dictionary learning on unstructured pre-treatment  
 51 data, and with downstream interpretability tools that map selected concepts back to human-  
 52 understandable summaries.

53 This formulation also suggests a concrete route to *interpretable heterogeneous treatment effect rep-*  
 54 *resentation learning*: first construct candidate concepts from complex pre-treatment observations,  
 55 then remove redundancy by testing whether a concept still carries treatment-effect heterogeneity in-  
 56 formation after accounting for already selected ones. This makes it possible to extend experimental  
 57 and quasi-experimental analyses beyond curated baseline covariates while retaining compactness  
 58 and interpretability.

59 Summing up, our contributions are as follows:

- 60 i. We formalize heterogeneous-treatment-effect identification from complex pre-treatment  
 61 observations as the problem of recovering one minimal sufficient subset of observed con-  
 62 cepts.
- 63 ii. We characterize the mismatch between representative recovery and vanilla coordinate-  
 64 wise multiple testing, and propose *Neural Effect Modifier Search* (NEMS), a recursive  
 65 conditional-testing procedure aimed at recovering compact representative sets rather than  
 66 all marginally active concepts.

- iii. We validate the framework on synthetic experiments and demonstrate its applied potential on real experimental and quasi-experimental analyses augmented with complex pre-treatment observations, yielding novel heterogeneity characterizations.

## 2 Latent Heterogeneous Treatment Effect Identification

We formalize here the population target and the corresponding observed-space recovery problem. **Note:** We discuss it for a randomized experiment with binary treatment, keeping the notation minimal while capturing the main ideas. The same development extends naturally to other causal heterogeneity functionals, such as treatment-on-the-treated heterogeneity in quasi-experimental settings.

### 2.1 Heterogeneous Treatment Effect within complex pre-treatment measurements

Consider an experiment with  $n$  units indexed by  $i \in [n] := \{1, \dots, n\}$ , a randomized binary intervention  $T_i \in \{0, 1\}$  (*treatment*), and a measured target of interest  $Y_i \in \mathbb{R}$  (*outcome*). Under the potential-outcomes framework [Rubin, 1974], each unit has potential outcomes  $Y_i(0), Y_i(1)$  defining the unit-level treatment effect  $\tau_i := Y_i(1) - Y_i(0)$ . Our goal is to characterize how the effect of  $T_i$  on  $Y_i$  varies across units, due to the interaction between  $T_i$  and a set of latent individual characteristics  $\mathbf{W}_i^{\text{dir}} \in \mathbb{R}^r$ , the *direct* effect modifiers [VanderWeele and Robins, 2007]. The population heterogeneity functional we wish to recover is the Conditional Average Treatment Effect (CATE):

$$\tau(\mathbf{w}) := \mathbb{E}[\tau_i \mid \mathbf{W}_i^{\text{dir}} = \mathbf{w}]. \quad (1)$$

Recovering  $\mathbf{W}_i^{\text{dir}}$  is desirable for two complementary reasons. First,  $\mathbf{W}_i^{\text{dir}}$  is a *minimal and sufficient* summary of the heterogeneity: by construction, all causal pathways from any other heterogeneity-relevant variable to  $\tau_i$  pass through  $\mathbf{W}_i^{\text{dir}}$ , which therefore d-separates  $\tau_i$  from the rest of the relevant latent space and provides a Markov blanket of  $\tau_i$  [Pearl, 1988, Koller and Friedman, 2009]. Second, direct modifiers are the only variables on which interventions [FL: directly] translate into changes of the treatment effect; recovering them yields a minimal characterization that is also *actionable* where intervention is feasible. [FL: an intervention on  $\mathbf{W}$  indirect also affects  $Y$ . I think we should say directly translate. A reviewer may complain about this, maybe we could also say that indirect effects can also be recursively found, which is why we focus on direct.]

The four-type effect modification taxonomy by VanderWeele and Robins [2007] clarifies what the rest of the heterogeneity-relevant latent space looks like (see Figure 3): besides direct modifiers  $\mathbf{W}^{\text{dir}}$ , one finds *indirect* modifiers  $\mathbf{W}^{\text{ind}}$  acting on  $\tau$  only through a direct one, *proxies*  $\mathbf{W}^{\text{prx}}$  descending from a direct modifier, and modifiers *by common cause*  $\mathbf{W}^{\text{cc}}$  sharing a common ancestor with a direct modifier. All four types display marginal effect modification, but only direct modifiers carry the causal handle on  $\tau$ : indirect modifiers act only mediated, while proxies and common-cause modifiers have no causal pathway to  $\tau$  at all, which should be controlled carefully. Indeed, consider an humanitarian program where market access is the true direct modifier but not directly measured, road density visible from satellite imagery may surface as a heterogeneity correlate, while most probably an effect modifier by common cause. A naive interpretation would suggest building roads to amplify the program effect, while the actual driver is market activity; a road built to a place with no market would not help. [FL: Super cool!]

In practice  $\mathbf{W}_i^{\text{dir}}$  is rarely fully measured directly, and is instead entangled in complex pre-treatment observations. Let  $\mathbf{X}_i \in \mathcal{X}$  denote the aggregation of all such pre-treatment measurements, possibly multimodal and combining structured covariates, e.g., questionnaire responses, with unstructured signals such as satellite imagery. We assume:

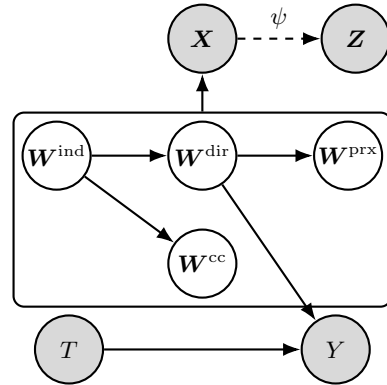


Figure 3: Causal model for latent heterogeneous treatment effect identification. Gray nodes are observed, white nodes are unobserved.

**Assumption 1** (Measurement Sufficiency). *The pre-treatment measurements  $\mathbf{X}_i$  carry all the (latent) effect-heterogeneity information, i.e.,*

$$\mathbb{E}[\tau_i \mid \mathbf{W}_i^{\text{dir}}] = \mathbb{E}[\tau_i \mid \mathbf{X}_i] \quad a.s. \quad (2)$$

Assumption 1 is plausible in well-designed experiments where the relevant factors of variation are controlled and otherwise the individual characteristics are extensively measured. It is also necessary: if violated, complete explanation of effect heterogeneity becomes impossible by design, and any analysis is forced to approximate  $\tau(\mathbf{w})$  using non-direct modifiers within  $\mathbf{X}_i$ , with potentially misleading interpretations. The causal heterogeneity functional is then identified at the  $\mathbf{X}$ -level by the controlled experiment design, and by Assumption 1 transferring to the conceptual target  $\tau(\mathbf{w})$ .

Existing methods for analyzing treatment-effect heterogeneity from  $\mathbf{X}_i$  either lack *expressivity* to handle (multimodal) complex measurements or sacrifice *interpretability*, among other limitations (see Section 5 for a full discussion). We overcome this trade-off, narrowing the application to controlled experiments with extensive individual measurements to empirically enforce Assumption 1 and leveraging modern representation learning to interpret such complex data recordings. To the best of our knowledge this work propose the first heterogeneous treatment effect interpretable characterization leveraging learned neural representation.

## 2.2 Learning interpretable representation for direct effect modifier identification

[FL: I wouldn't cite the platonic hypothesis, there are several papers claiming it's false. I would say that linear probing on top of foundation models is increasingly more performant, meaning that information is linearly accessible in the features.] In the spirit of the *Platonic Representation Hypothesis* [Huh et al., 2024], we conjecture that pretrained foundation models, trained on extensive corpora across modalities, spontaneously surface representations whose coordinates approximately disentangle the latent factors of variation of the world [not true, their are sufficient but very entangled.. the SAE make the disentanglement], including the direct modifiers we aim to identify [FL: reasonable argument? better refs?]. Rather than learning [from scratch] a representation tailored to the heterogeneity task on the experimental data alone, which would not be feasible at experimental sample sizes, we propose to extract a pre-treatment measurements representation

$$\mathbf{Z}_i := \psi(\mathbf{X}_i) \in \mathbb{R}^m \quad (3)$$

from such a pretrained model and identify, among its coordinates, those aligned with  $\mathbf{W}^{\text{dir}}$ . A natural choice for  $\psi$  is a sparse autoencoder trained on top of a foundation-model encoder for the modality of interest [Cunningham et al., 2023, Bricken et al., 2023], optionally combined with classical dictionary-learning components [Mairal et al., 2009] and appended hand-crafted covariates (if available). Sparse autoencoders empirically concentrate each concept onto a dominant, distinct dictionary coordinate [Makelov et al., 2024, Pach et al., 2025], the regime our identification objective exploits.

**Assumption 2** (Principal Alignment). *There exists a unique injective map  $\pi : [r] \hookrightarrow [m]$ ,  $k \mapsto j_k$ , such that for every  $k \in [r]$ :*

$$\mathbf{W}_i^{\text{dir},k} \perp\!\!\!\perp \mathbf{Z}_i^{[m] \setminus \{j_k\}} \mid \mathbf{Z}_i^{j_k}. \quad (4)$$

*We call  $\mathcal{S}^* := \{j_1, \dots, j_r\}$  the set of neural exposure interactions coordinates, and  $\mathbf{Z}_i^{j_k}$  the neural exposure interactions for  $\mathbf{W}_i^{\text{dir},k}$ .*

Principal Alignment asks that each direct modifier be summarized by one dominant coordinate, with no residual concept-specific signal scattered across the others [FL: you could cite the third pillar paper here? Condition seems highly related]. It is a population-level idealization of the regime SAEs aim for, and whether it holds for a specific dictionary can be probed empirically for supervised concepts (Appendix C). Under Assumption 2,  $\mathcal{S}^*$  is well-defined as the unique set of dominant coordinates encoding  $\mathbf{W}^{\text{dir}}$ ; we develop the recoverability of  $\mathcal{S}^*$  from observed data in Section 4.

## 154 2.3 Actionable Heterogeneous Treatment Effect Identification

$$\text{Task. } X \xrightarrow{\psi} Z \xrightarrow{\text{NEIS}} Z^{S^*} \approx W^{\text{dir}}$$

*Recover, from a single controlled experiment satisfying Measurement Sufficiency (A1) and Principal Alignment (A2), the representation coordinates encoding the direct effect modifiers of  $T$  on  $Y$ .  
[potentially replace with pipeline summary but could be redundant with the teaser]*

155

156 [RC: contents is already here but rewrite/structure paragraphs][FL: I think here you want to stress  
157 that just doing a SAE on  $X$  doesn't return the direct modifiers but finds everything. It's roughly al-  
158 ready in the text, but needs to be super clear. Consider putting a **Challenge**: somewhere to highlight]  
159 The representation  $Z$  encodes all heterogeneity-relevant variables in  $X$ , including direct, indirect,  
160 proxy, and common-cause effect modifiers (Figure 3); identifying  $S^*$  requires isolating only the  
161 direct ones. Principal Alignment constrains the conditional structure linking each  $W^{\text{dir},k}$  to its  
162 dominant coordinate, but it does not constrain marginal dependences between  $W^{\text{dir},k}$  and the many  
163 proxy, indirect, or common-cause coordinates that share information with it. As a result, vanilla  
164 effect-modifier search returns coordinates that are typically more numerous and marginally indis-  
165 distinguishable from neural exposure interactions coordinates, yielding an uninterpretable summary.  
166 Beyond formal feasibility, interpretable identification is also *useful* under a parsimony prior con-  
167 sistent with empirical experience: in physical and social systems, treatment-effect heterogeneity is  
168 typically driven by a small number of substantively meaningful mechanisms, even when the pre-  
169 treatment measurement is high-dimensional. We do not claim parsimony as an assumption but as  
170 the empirical regime under which interpretable identification is both possible and useful.

171 Target the standalone direct effect modifiers supports two distinct downstream uses. First, being a  
172 minimal sufficient set of covariates summarizing the full effect heterogeneity they can be plugged  
173 into any interpretable HTE estimator to produce subgroup-level effect estimates on the identified  
174 mechanism. Second, by definition, they are candidate policy levers: under additional standard causal  
175 assumptions, including interventional consistency on  $W^{\text{dir}}$  [All: check] and absence of internal  
176 causation within  $W^{\text{dir}}$ , modifications of these factors translate into corresponding changes of the  
177 treatment effect. These additional assumptions are generally not testable from the experimental data  
178 alone, and policy-level claims should be made grain of salt without further experimentation.

179 The remainder of the paper develops a procedure for  $S^*$  identification. Section 3 characterizes  
180 vanilla effect-modifier search and shows that, by design, it returns a marginally-active superset of  $S^*$   
181 that is neither interpretable as a heterogeneity characterization nor translatable into policy. Section 4  
182 introduces NEIS, our iterative procedure to recover  $S^*$  directly.

## 183 3 From Effect Modification to Interaction

184 A natural baseline for identifying the latent heterogeneous treatment effect over the learned repre-  
185 sentation is coordinate-wise multiple testing for effect modification. However, in learned represen-  
186 tations even principally aligned dictionaries are empirically broadly entangled [Chanin et al., 2024],  
187 so the great majority of marginally active coordinates for effect modification are *non-direct* effect  
188 modifiers. Let

$$\mathcal{M}^* := \left\{ j \in [m] : \mathbb{P}\left(\mathbb{E}[\tau_i | Z_i^j] \neq \mathbb{E}[\tau_i]\right) > 0 \right\}, \quad (5)$$

189 the set of coordinates marginally active for effect modification, then, each underlying direct modifier  
190 is reflected, even if minimally, in many candidate  $\mathcal{M}^*$  coordinates as proxies, indirect descendants,  
191 or common-cause neighbors, while only one of them is the principal identifier.<sup>1</sup> The natural regime  
192 in learned representations is therefore

$$|S^*| \ll |\mathcal{M}^*|, \quad (6)$$

193 making  $\mathcal{M}^*$  significantly redundant, uninterpretable and detached from any causal interpretation,  
194 i.e., hiding true interactions within overly redundant effect modification factors [VanderWeele,

<sup>1</sup>A principally aligned coordinate could in principle fail to be marginally active under a Simpson-style cancellation, where the direct-modifier signal is masked at the marginal level and surfaces only conditionally. For simplicity of discussion we do ignore here such pathological regime, i.e., faithfulness [Spirtes et al., 2000], but still address it in our mitigation [FL: reasonable argument?]

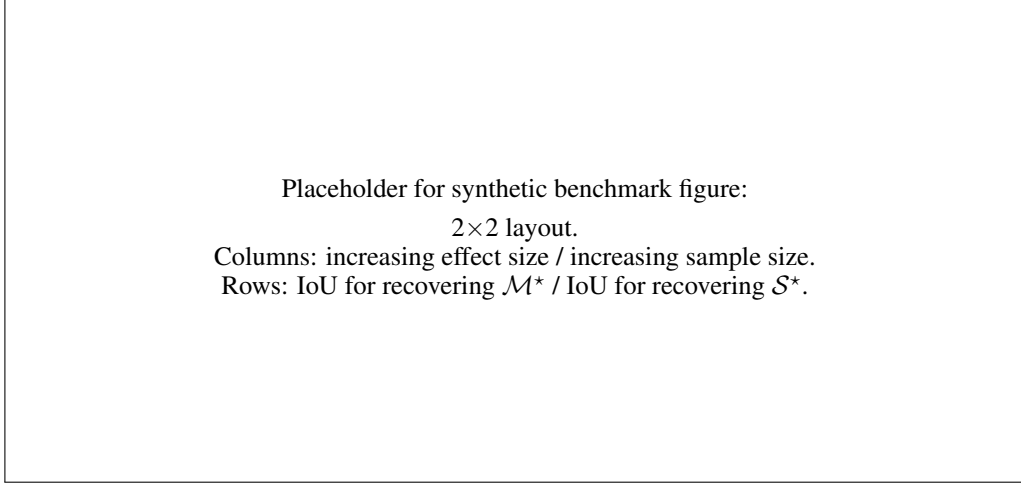


Figure 4: Synthetic benchmark illustrating the experimental power paradox. The top row reports recovery of the marginal-activity set  $\mathcal{M}^*$ ; the bottom row reports recovery of the principal-proxy set  $\mathcal{S}^*$ . Columns vary effect size (left) and sample size (right). Only two coordinates form  $\mathcal{S}^*$ , while many others are non-direct effect modifiers marginally active on  $\tau$ . As signal strength or sample size increases, marginal screening improves recovery of  $\mathcal{M}^*$  but remains misaligned with the interpretable target  $\mathcal{S}^*$ ; NEMS is designed to track the latter directly.

2009]. Even heuristic selection by marginal activation magnitude does not resolve the problem: a strongly-activating proxy of a high-magnitude direct modifier can outrank a low-magnitude direct modifier, scrambling any meaningful ordering, and the cutoff between principal and non-principal coordinates is unknown a priori.

**Paradox.** As experimental power grows, with sample size, signal strength, or even test sensitivity, marginal screening converges to its own target  $\mathcal{M}^*$  and reports an increasing number of non-direct modifiers, each backed by genuinely significant evidence. The procedure is consistent for what it tests, still producing a mismatch between statistical and scientific success. Indeed, the scientific target for interpretable and useful latent heterogeneous treatment effect identification is  $\mathcal{S}^*$ , not  $\mathcal{M}^*$ . More power makes the mismatch more visible, not less.

**Empirical evidence.** We illustrate this mismatch with a semi-synthetic experiment based on CelebA [Liu et al., 2018]. We construct a randomized controlled trial in which the true direct modifiers are two visual attributes, *wearing sunglasses* and *wearing a hat*. The pre-treatment information consists only of face images, with no annotated covariates. Candidate concepts are learned by a sparse autoencoder with  $m = 5,000$  dictionary elements on top of SigLIP representations [Zhai et al., 2023]; full details are given in Appendix C. We report results in Figure 4.

[discuss results once figure is in. additional interpret a cherry picked false discovery to illustrate the multiple testing fallacy]

## 4 Neural Exposure Interaction Search

We propose *Neural Exposure Interaction Search* (NEIS), an iterative procedure for identifying neural exposure interactions coordinates from a single experimental sample. At each round NEIS adds the candidate coordinate that, conditional on what is already selected, carries the strongest residual heterogeneity signal, and removes any previously-selected coordinate that has become redundant given the rest. The atomic operation is a *conditional CATE-independence test*, returning, for a candidate  $j \in [m]$  and a current selection  $S \subseteq [m] \setminus \{j\}$ , a  $p$ -value  $p_j(S)$  for

$$H_0(j | S) : \mathbb{E}[\tau_i | \mathbf{Z}_i^{S \cup \{j\}}] = \mathbb{E}[\tau_i | \mathbf{Z}_i^S] \quad \text{a.s.} \quad (7)$$

Concrete instantiations and practical defaults are detailed in Appendix B.

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### Algorithm 1 Neural Exposure Interaction Search (NEIS)

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1: Input:  $\{(\mathbf{Z}_i, T_i, Y_i)\}_{i=1}^n$ , significance level  $\alpha$ 
2: Output:  $S \subseteq [m]$ 
3:  $S \leftarrow \emptyset$ ,  $S_{\text{prev}} \leftarrow \{-1\}$ 
4: while  $S \neq S_{\text{prev}}$  do
5:    $S_{\text{prev}} \leftarrow S$ ,  $\bar{S} \leftarrow [m] \setminus S$ 
6:    $j^* \leftarrow \arg \min_{j \in \bar{S}} p_j(S)$  ▷ forward step
7:   if  $p_{j^*}(S) \leq \alpha/|\bar{S}|$  then  $S \leftarrow S \cup \{j^*\}$ 
8:   end if
9:   for  $j \in S$  do ▷ backward step
10:    if  $p_j(S \setminus \{j\}) > \alpha/|S|$  then  $S \leftarrow S \setminus \{j\}$ 
11:    end if
12:   end for
13: end while

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**Discussion** NEIS adapts the classic forward-backward Markov-blanket (MB) discovery skeleton [Tsamardinos et al., 2003, Aliferis et al., 2010] to a new target and a new setting: it discovers the Markov blanket of the CATE functional, which is not directly observed, on top of a learned representation of complex pre-treatment observations rather than hand-crafted structured covariates. The challenge is that classical MB discovery, applied directly to  $\mathbf{Z}$ , recovers *some* Markov blanket of the target functional but not necessarily one corresponding to direct effect modifiers: with  $m \gg r$  and pervasive concept overlap, multiple sufficient subsets typically exist, and standard guarantees do not select the interpretable, actionable one. Principal Alignment (Assumption 2) in modern representation learning is what closes this gap: it makes the principal-proxy set  $\mathcal{S}^*$  the unique minimal sufficient subset, and turns MB discovery into direct-modifier identification on unstructured pre-treatment observations. Forward growth ensures every direct modifier eventually enters the selection; backward pruning re-tests retained coordinates against the full current selection at each round, removing any that has become redundant.

**Theorem 4.1** (Selection Consistency). *Let  $\hat{\mathcal{S}}_n$  denote the output of NEIS on a sample of size  $n$  at fixed level  $\alpha \in (0, 1)$ . Under Measurement Sufficiency (Assumption 1), Principal Alignment (Assumption 2), faithfulness, and validity of the conditional CATE-independence test,*

$$\mathbb{P}(\underbrace{\mathcal{S}^* \subseteq \hat{\mathcal{S}}_n}_{\text{recall}=1}) \rightarrow 1 \quad \text{as } n \rightarrow \infty \quad \text{and} \quad \mathbb{P}(\underbrace{\hat{\mathcal{S}}_n \subseteq \mathcal{S}^*}_{\text{precision}=1}) \geq 1 - \alpha \quad \text{for all } n.$$

*Proof sketch.* Principal Alignment combined with Measurement Sufficiency makes  $\mathcal{S}^*$  the unique minimal subset of  $\mathbf{Z}$  sufficient for  $\mathbb{E}[\tau_i | \mathbf{Z}_i]$ , so identification reduces to Markov-blanket discovery on the CATE; the forward-backward selection then delivers asymptotic recall (every direct modifier wins a forward step under test consistency) and finite-sample precision (every retained coordinate passes a Bonferroni-corrected backward check at termination, controlled by union bound). The full development is in Appendix A.  $\square$



The two guarantees are asymmetric by design. Recall is asymptotic because it relies on the test’s consistency against fixed alternatives, available only in the large-sample limit. Precision is finite-sample because the false-discovery probability is controlled by the Bonferroni gate at termination, regardless of  $n$  and it strictly depends on the backward step: without it, a coordinate that passed an early forward gate under partial conditioning would persist, leaving precision unattainable at fixed  $(\alpha)$ . Together, they imply  $\liminf_n \mathbb{P}(\hat{\mathcal{S}}_n = \mathcal{S}^*) \geq 1 - \alpha$ , the fixed- $\alpha$  analogue of strict selection consistency  $\mathbb{P}(\hat{\mathcal{S}}_n = \mathcal{S}^*) \rightarrow 1$ , which would additionally require  $\alpha_n \rightarrow 0$ .

## 5 Related Work

**Heterogeneous treatment effect identification.** Existing HTE identification methods trade expressivity and required human supervision with interpretability and downstream policy relevance. *Decision rule*-based methods [Foster et al., 2011, Athey and Imbens, 2016, Bargagli-Stoffi et al., 2020] return interpretable subgroups but operate on expensive and potentially limited hand-crafted covariates. *Forest*-based estimators [Wager and Athey, 2018, Athey et al., 2019, Hahn et al., 2020] estimate  $\mathbb{E}[\tau_i | \mathbf{X}_i]$  flexibly on structured  $\mathbf{X}$ , but return pointwise predictions rather than identified modifiers and require the analyst to provide semantically interpretable covariates upfront. *Meta-learners* [Künzel et al., 2019, Nie and Wager, 2021, Kennedy, 2023] are agnostic to the base estimator and can in principle wrap pretrained foundation-model nuisances on unstructured  $\mathbf{X}$ , but they have not been demonstrated to do so at experimental scale, and they share the same uninterpretable pointwise output. *Post-hoc* CATE interpretation methods [Hines et al., 2022, Paillard et al., 2024] sit on top of any base estimator and rank candidate modifiers by importance, partially recovering interpretability, but inherit the base estimator’s covariate constraints and offer no guarantee that highly ranked variables are direct effect modifiers. *Deep Learning*-based estimators [Jerzak et al., 2023] extend HTE estimation to unstructured measurements such as imagery, gaining expressivity at the cost of modifier-level interpretability. Table 1 positions NEMS against these families. NEMS achieves all the desiderata by leveraging modern experimental practice (extensive measurement collections supporting Measurement Sufficiency) and modern representation learning (sparse-autoencoder-based dictionaries supporting Representation Sufficiency and Principal Alignment).

Table 1: HTE-method properties. *Empiricist*: automated feature extractions from raw pre-treatment observations. *Scaling complexity*: handling unstructured / multimodal pre-treatment observations. *Interpretable*: human readable HTE characterization. *Actionable*: targets treatment-interactive factors.

| Method                      | Empiricist    | Scaling complexity | Interpretable | Actionable |
|-----------------------------|---------------|--------------------|---------------|------------|
| <i>Decision rule</i> -based | ✗             | ✗                  | ✓             | ✗          |
| <i>Forest</i> -based        | ✗             | ✗                  | ✗             | ✗          |
| Meta-Learners               | (potentially) | (potentially)      | ✗             | ✗          |
| Post-hoc interpretations    | ✗             | ✗                  | ✓             | ✗          |
| <i>Deep Learning</i> -based | ✓             | ✓                  | ✗             | ✗          |
| <b>Ours</b>                 | ✓             | ✓                  | ✓             | ✓          |

**Representation learning for causal downstream tasks.** Causal Representation Learning was first articulated as the recovery of causally meaningful latent variables from raw observations as a general-purpose target [Schölkopf et al., 2021], an ambitious goal that faces fundamental identifiability barriers. The field has since moved toward well-specified causal targets, in the spirit already suggested by Causal Feature Learning [Chalupka et al., 2017] albeit in controlled, low-dimensional settings. Pretrained encoders are now integrated into specific causal pipelines: for treatment-effect estimation on unstructured outcomes [Cadei et al., 2024, 2025], for unsupervised effect discovery on unstructured outcomes [Mencattini et al., 2025], for confounding adjustment via text embeddings [Veitch et al., 2020], and for HTE estimation on imagery [Jerzak et al., 2023]. Our work extends this empiricist line, unlocking, at least in the attempt, the identification of an interpretable and actionable HTE characterization from unstructured *pre-treatment* observations.

**Dictionary learning and interpretability.** Foundation-model representations admit several routes to a candidate concept dictionary: classical sparse coding [Mairal et al., 2009], sparse autoencoders



280 [Cunningham et al., 2023, Bricken et al., 2023, Gao et al., 2024], and recent variants addressing  
281 entanglement failure modes such as feature absorption [Chanin et al., 2024], including transcoders  
282 [Dunefsky et al., 2024] and Matryoshka SAEs [Bussmann et al., 2025]. Once a concept is identified,  
283 an interpretation pipeline translates it into a human-readable summary via top-activating examples,  
284 supervised probes, or LM-assisted descriptions of activating inputs [Bills et al., 2023]. NEMS is  
285 agnostic to the specific dictionary and interpretation pipeline; we use SAEs on top of foundation-  
286 model encoders as a concrete instantiation, with VLM-assisted summaries contrasting top-activating  
287 against random-activation inputs.

## 288 **6 Extending humanitarian programmes interpretation**

289 [Showcase significance of our method. First we extend existing humanitarian programme with ad-  
290 ditional raw measuremnts (unprocessable before). We then identify new heterogeneity drivers, better  
291 charcetherizing programme impact, hopefully leading to improvement in the next iterations.]

### 292 **6.1 Application 1: Youth Opportunities Program (Uganda)**

293 **Experiment design.**

294 **Analysis.**

295 **Results.**

### 296 **6.2 Apprlication 2: LEAP 1000 Programme (Ghana)**

297 **Experiment design.**

298 **Analysis.**

299 **Results.**

## 300 **7 Conclusion**

## References

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## 787 A Proof of Theorem 4.1

788 We establish that NEIS recovers  $S^*$  with asymptotic recall and finite-sample precision under Mea-  
789 surement Sufficiency, Principal Alignment, faithfulness, and validity of the conditional CATE-  
790 independence test. The argument has two stages:

- 791 i. Principal Alignment combined with Measurement Sufficiency makes  $S^*$  the unique mini-  
792 mal subset of  $Z$  sufficient for  $\mathbb{E}[\tau_i | Z_i]$ , reducing identification to Markov-blanket discov-  
793 ery on the CATE (Lemmas A.1–A.2).
- 794 ii. A standard IAMB-style argument [Tsamardinos et al., 2003] on this Markov blanket yields  
795 the recovery guarantees (Theorem 4.1).

### 796 A.1 Markov-blanket structure

**Lemma A.1** (Joint conditional independence). *Under Principal Alignment,*

$$W_i^{\text{dir}} \perp\!\!\!\perp Z_i^{[m] \setminus S^*} \mid Z_i^{S^*}. \quad (8)$$

797 *Proof.* Principal Alignment provides, for each  $k \in [r]$ ,  $W_i^{\text{dir},k} \perp\!\!\!\perp Z_i^{[m] \setminus \{j_k\}} \mid Z_i^{j_k}$ . By the contrac-  
798 tion property of conditional independence, since  $S^* \supseteq \{j_k\}$  and  $Z^{[m] \setminus S^*} \subseteq Z^{[m] \setminus \{j_k\}}$ ,

$$W_i^{\text{dir},k} \perp\!\!\!\perp Z_i^{[m] \setminus S^*} \mid Z_i^{S^*}. \quad (9)$$

799 Combining across  $k \in [r]$  using the injectivity of  $\pi$  and the conditional-independence chain rule  
800 yields Eq. (8).  $\square$

**Lemma A.2** (CATE Markov-blanket structure). *Under Measurement Sufficiency and Principal Alignment,*

$$\mathbb{E}[\tau_i | Z_i] = \mathbb{E}[\tau_i | Z_i^{S^*}] \quad \text{a.s.}, \quad (10)$$

*and  $S^*$  is the unique minimal subset of  $[m]$  with this property.*

801 *Proof.* By the tower property,

$$\mathbb{E}[\tau_i | Z_i] = \mathbb{E}[\mathbb{E}[\tau_i | W_i^{\text{dir}}, Z_i] | Z_i]. \quad (11)$$

802 Since  $Z_i = \psi(X_i)$  is a deterministic function of  $X_i$ , and by Measurement Sufficiency  $\mathbb{E}[\tau_i | X_i] =$   
803  $\mathbb{E}[\tau_i | W_i^{\text{dir}}]$ , we have  $\mathbb{E}[\tau_i | W_i^{\text{dir}}, Z_i] = \mathbb{E}[\tau_i | W_i^{\text{dir}}]$  a.s. Therefore

$$\mathbb{E}[\tau_i | Z_i] = \mathbb{E}[\mathbb{E}[\tau_i | W_i^{\text{dir}}] | Z_i]. \quad (12)$$

804 By Lemma A.1, the conditional distribution of  $W_i^{\text{dir}}$  given  $Z_i$  coincides with its conditional distri-  
805 bution given  $Z_i^{S^*}$  alone. Hence

$$\mathbb{E}[\mathbb{E}[\tau_i | W_i^{\text{dir}}] | Z_i] = \mathbb{E}[\mathbb{E}[\tau_i | W_i^{\text{dir}}] | Z_i^{S^*}] = \mathbb{E}[\tau_i | Z_i^{S^*}], \quad (13)$$

806 where the last equality applies the same tower argument to  $Z_i^{S^*}$ . Eq. (10) follows.

807 *Minimality.* Suppose  $S \subsetneq S^*$  and pick  $j_k \in S^* \setminus S$ . By Lemma A.1,  $Z_i^{j_k}$  is the only coordinate  
808 that, conditionally, exhausts the information of  $W_i^{\text{dir},k}$ ; since  $j_k \notin S$ , the conditional distribution of  
809  $W_i^{\text{dir},k}$  given  $Z_i^S$  differs from that given  $Z_i^{S \cup \{j_k\}}$ , so  $\mathbb{E}[\tau_i | Z_i^S] \neq \mathbb{E}[\tau_i | Z_i]$  on a set of positive  
810 probability. Hence  $S$  is not sufficient.

811 *Uniqueness.* Suppose  $S'$  is another minimal sufficient subset. Minimality forces  $|S'| = r$ , and for  
812 each  $k \in [r]$ ,  $S'$  must contain a coordinate that captures  $W_i^{\text{dir},k}$  up to conditional independence; by  
813 Principal Alignment the only such coordinate is  $j_k$ . Hence  $S' = S^*$ .  $\square$

## 814 A.2 Selection Consistency

815 The recovery argument requires two regularity additional properties: validity of the conditional  
816 CATE-independence test and faithfulness of the joint distribution.

**Assumption 3** (Test validity and consistency). *For any fixed  $S$  and  $j$ : under  $H_0(j | S)$ ,  $p_j(S)$  is asymptotically uniform on  $[0, 1]$ ; under any fixed alternative,  $\mathbb{P}(p_j(S) < t) \rightarrow 1$  as  $n \rightarrow \infty$  for any  $t > 0$ .*

**Assumption 4** (Faithfulness). *The joint distribution of  $(\tau_i, \mathbf{Z}_i)$  is faithful to the generating structure:  $H_0(j | S)$  holds at the population level if and only if  $j$  is structurally conditionally independent of  $\tau_i$  given  $\mathbf{Z}_i^S$  in the underlying DAG.*

**Theorem** (Selection Consistency). *Let  $\hat{S}_n$  denote the output of NEIS on a sample of size  $n$  at fixed level  $\alpha \in (0, 1)$ . Under Measurement Sufficiency, Principal Alignment, Test validity (Assumption 3), and Faithfulness (Assumption 4),*

$$\mathbb{P}(\mathcal{S}^* \subseteq \hat{S}_n) \rightarrow 1 \quad \text{as } n \rightarrow \infty \quad \text{and} \quad \mathbb{P}(\hat{S}_n \subseteq \mathcal{S}^*) \geq 1 - \alpha \quad \text{for all } n.$$

817 *Proof.* We prove the Recall and Precision result distinctly.

818 *Recall.* Fix  $k \in [r]$  and consider any round at which  $j_k \notin S$ , the current selection. By Lemma A.2,  
819  $\mathbb{E}[\tau_i | \mathbf{Z}_i^{S \cup \{j_k\}}] \neq \mathbb{E}[\tau_i | \mathbf{Z}_i^S]$  on a set of positive probability, so  $H_0(j_k | S)$  is false. By Assump-  
820 tion 3,  $\mathbb{P}(p_{j_k}(S) < \alpha/|[m] \setminus S|) \rightarrow 1$ . For non-principal coordinates  $j' \in [m] \setminus \mathcal{S}^*$ , Lemma A.1  
821 and Assumption 4 imply that  $H_0(j' | S)$  holds whenever  $\mathcal{S}^* \subseteq S$ ; under any partial selection, non-  
822 principal coordinates either have true nulls (asymptotically uniform  $p$ -values) or true alternatives  
823 that are dominated by  $p_{j_k}(S)$  at the population level. Hence  $j_k$  achieves the smallest  $p$ -value among  
824 unselected coordinates and passes the gate with probability  $\rightarrow 1$ , and is added at some forward step.  
825 Once added,  $j_k$  is removed at a backward step only if  $p_{j_k}(\hat{S} \setminus \{j_k\}) > \alpha/|\hat{S}|$ ; by Lemma A.2 this  
826 corresponds to a true alternative, so by Assumption 3 the probability of failing the threshold tends  
827 to zero. Iterating across  $k \in [r]$  gives  $\mathbb{P}(\mathcal{S}^* \subseteq \hat{S}_n) \rightarrow 1$ .

828 *Precision.* At termination, every  $j \in \hat{S}_n$  has passed the final backward check:  $p_j(\hat{S}_n \setminus \{j\}) \leq$   
829  $\alpha/|\hat{S}_n|$ . Suppose  $j' \in \hat{S}_n \setminus \mathcal{S}^*$ . By Lemma A.2 and Assumption 4,  $H_0(j' | \hat{S}_n \setminus \{j'\})$  holds at the  
830 population level. By Assumption 3,  $p_{j'}(\hat{S}_n \setminus \{j'\})$  is uniform on  $[0, 1]$ , so

$$\mathbb{P}(p_{j'}(\hat{S}_n \setminus \{j'\}) \leq \alpha/|\hat{S}_n|) \leq \alpha/|\hat{S}_n|. \quad (14)$$

831 A union bound over the at most  $|\hat{S}_n|$  candidate spurious coordinates yields  $\mathbb{P}(\hat{S}_n \not\subseteq \mathcal{S}^*) \leq \alpha$ , so  
832  $\mathbb{P}(\hat{S}_n \subseteq \mathcal{S}^*) \geq 1 - \alpha$  for all  $n$ .  $\square$

833 *Note:* Combining recall and precision, one obtains  $\liminf_n \mathbb{P}(\hat{S}_n = \mathcal{S}^*) \geq 1 - \alpha$ . With a vanishing  
834 schedule  $\alpha_n \rightarrow 0$  such that  $\alpha_n \cdot |\hat{S}_n| \rightarrow 0$ , the precision bound becomes  $\mathbb{P}(\hat{S}_n \subseteq \mathcal{S}^*) \rightarrow 1$ ,  
835 and strict selection consistency  $\mathbb{P}(\hat{S}_n = \mathcal{S}^*) \rightarrow 1$  follows. We adopt the fixed- $\alpha$  form because it  
836 matches the algorithm as run in practice; it does not require tuning a vanishing schedule and gives a  
837 non-asymptotic precision guarantee.

## 838 B Method details

### 839 B.1 Conditional CATE-independence test

840 NEIS reduces to a sequence of conditional CATE-independence tests: for any candidate  $j \in [m]$   
841 and current selection  $S \subseteq [m] \setminus \{j\}$ , produce a  $p$ -value  $p_j(S)$  for

$$H_0(j | S) : \quad \mathbb{E}[\tau_i | \mathbf{Z}_i^{S \cup \{j\}}] = \mathbb{E}[\tau_i | \mathbf{Z}_i^S] \quad \text{a.s.} \quad (15)$$

842 The recovery guarantee of Theorem 4.1 requires only that  $p_j(\cdot)$  be asymptotically uniform under the  
 843 null and consistent against fixed alternatives (Assumption 3); any test family meeting this require-  
 844 ment can be plugged into the procedure. Since  $\tau_i$  is unobserved, the test cannot be applied directly  
 845 to  $(\tau_i, \mathbf{Z}_i)$ ; instead, we construct a pseudo-outcome  $\hat{\varphi}_i$  satisfying

$$\mathbb{E}[\hat{\varphi}_i \mid \mathbf{Z}_i] = \mathbb{E}[\tau_i \mid \mathbf{Z}_i] \quad \text{a.s.} \quad (16)$$

846 and apply the test to  $(\hat{\varphi}_i, \mathbf{Z}_i)$ . We describe a doubly-robust default implementation that is robust to  
 847 nonlinear heterogeneity, and a simpler linear baseline used for comparison.

848 **Doubly-robust GCM test (default).** For a randomized experiment with binary treatment, the stan-  
 849 dard pseudo-outcome is the R-learner residualization [Robinson, 1988, Nie and Wager, 2021]:

$$\hat{\varphi}_i = \frac{(Y_i - \hat{m}(\mathbf{Z}_i))(T_i - e)}{e(1 - e)}, \quad (17)$$

850 where  $e := \mathbb{P}(T_i = 1)$  is known by experimental design, and  $\hat{m}(\mathbf{Z}_i) := \hat{\mathbb{E}}[Y_i \mid \mathbf{Z}_i]$  is a cross-fitted  
 851 nuisance estimator (in practice, a ridge regression or small MLP on  $\mathbf{Z}_i$ , trained with  $K$ -fold cross-  
 852 fitting to ensure independence between the nuisance estimate and the residual at each unit). The  
 853 DR-learner pseudo-outcome of Kennedy [2023] is an alternative drop-in with the same property.

854 Given  $\hat{\varphi}_i$ , the conditional independence between  $\hat{\varphi}_i$  and  $Z_i^j$  given  $\mathbf{Z}_i^S$  is tested via the Generalised  
 855 Covariance Measure (GCM) of Shah and Peters [2020]. The GCM regresses  $\hat{\varphi}_i$  on  $\mathbf{Z}_i^S$  and  $Z_i^j$  on  
 856  $\mathbf{Z}_i^S$ , forms the per-unit residual product

$$R_i(j \mid S) := (\hat{\varphi}_i - \hat{\mathbb{E}}[\hat{\varphi}_i \mid \mathbf{Z}_i^S]) \cdot (Z_i^j - \hat{\mathbb{E}}[Z_i^j \mid \mathbf{Z}_i^S]), \quad (18)$$

857 and tests whether  $\mathbb{E}[R_i(j \mid S)] = 0$  via a normalized  $z$ -statistic

$$T_n(j \mid S) = \frac{\sqrt{n} \bar{R}_n(j \mid S)}{\hat{\sigma}_n(j \mid S)}, \quad (19)$$

858 where  $\bar{R}_n$  and  $\hat{\sigma}_n$  are the sample mean and sample standard deviation of  $R_i(j \mid S)$ . Under  
 859  $H_0(j \mid S)$ ,  $T_n(j \mid S)$  is asymptotically standard normal provided the two nuisance regressions are  
 860 consistent at rates whose product is  $o(n^{-1/2})$  — substantially weaker than parametric correctness  
 861 — and the test is consistent against any fixed conditional-mean alternative [Shah and Peters, 2020].  
 862 The resulting  $p$ -value is robust to nonlinear heterogeneity in  $\mathbf{Z}$  and yields the validity assumed by  
 863 Theorem 4.1.

864 **Linear treatment-interaction test (baseline).** A simpler alternative we consider as a baseline is  
 865 a linear treatment-interaction regression. Fit the working model

$$Y_i = \alpha + T_i \beta_T + \mathbf{Z}_i^S \gamma_S + T_i \cdot \mathbf{Z}_i^S \boldsymbol{\delta}_S + Z_i^j \beta_j + T_i \cdot Z_i^j \delta_j + \varepsilon_i, \quad (20)$$

866 and test  $H_0(j \mid S)$  via the partial  $F$ -test on  $(\beta_j, \delta_j)$ . This is fast, requires no cross-fitting, and  
 867 is straightforward to implement, but is only consistent against alternatives within the linear-in- $\mathbf{Z}$   
 868 class: heterogeneity that is genuinely nonlinear in  $Z^j$  given  $\mathbf{Z}^S$  (e.g., a threshold or U-shape) can be  
 869 missed at any sample size. Under correctly-specified linearity, Assumption 3 is satisfied; in learned-  
 870 representation settings where  $\mathbf{Z}$  is the output of a sparse autoencoder, the linearity assumption is  
 871 generally not warranted. We report results under both tests in Appendix C and use the doubly-robust  
 872 GCM test as the default in our applied analyses.

## 873 B.2 Minimal Python Implementation Snippet

874 We provide a minimal Python snippet of NEIS relying on pandas [McKinney et al., 2011] for  
 875 tabular operations, NumPy [Harris et al., 2020] and SciPy [Virtanen et al., 2020] for numerical  
 876 and statistical routines, statsmodels [Seabold et al., 2010] for the linear test, and scikit-learn  
 877 [Pedregosa et al., 2011] for nuisance regression in the doubly-robust variant. NEIS is parameterized  
 878 by a generic conditional CATE-independence test `cci_test`; we provide both the linear treatment-  
 879 interaction baseline and the doubly-robust GCM default of Appendix B.1.



880 **Neural Exposure Interaction Search** Iterative forward-backward selection until a fixed point is  
881 reached.

```

882 def NEIS(T, Y, Z, alpha=0.05, cci_test=None):
883     S, S_prev = [], None
884     while S != S_prev:
885         S_prev = list(S)
886         # forward step
887         bar_S = [c for c in Z.columns if c not in S]
888         tests = cci_test(T, Y, Z, S, candidates=bar_S)
889         j_star = tests["p_value"].idxmin()
890         if tests.loc[j_star, "p_value"] <= alpha / len(bar_S):
891             S.append(j_star)
892         # backward step
893         for j in list(S):
894             tests = cci_test(T, Y, Z, [c for c in S if c != j],
895                             candidates=[j])
896             if tests.loc[j, "p_value"] > alpha / len(S):
897                 S.remove(j)
898     return S

```

899 **Doubly-robust GCM test (default)** Constructs the R-learner pseudo-outcome via cross-fitted nuisance regression on  $Z$  (Eq. (17)), residualizes the candidate  $Z^j$  on  $Z^S$ , and applies the Generalised Covariance Measure of Shah and Peters [2020].

```

902 import numpy as np, pandas as pd, scipy.stats
903 from sklearn.ensemble import RandomForestRegressor
904 from sklearn.model_selection import KFold
905 def _crossfit(X, y, n_splits=5):
906     pred = np.zeros_like(y, dtype=float)
907     for tr, te in KFold(n_splits=n_splits, shuffle=True, random_state
908                        =0).split(X):
909         m = RandomForestRegressor(n_estimators=100, n_jobs=-1).fit(X[
910             tr], y[tr])
911         pred[te] = m.predict(X[te])
912     return pred
913 def cci_test_gcm(T, Y, Z, S, candidates):
914     e = T.mean()
915     Z_S = Z[S].to_numpy() if S else np.zeros((len(Y), 1))
916     # R-learner residualization on the current selection
917     m_hat = _crossfit(Z_S, Y.to_numpy())
918     phi = (Y.to_numpy() - m_hat) * (T.to_numpy() - e) / (e * (1 - e))
919     phi_resid = phi - _crossfit(Z_S, phi)
920     out = []
921     for j in candidates:
922         z_j = Z[j].to_numpy()
923         z_j_resid = z_j - _crossfit(Z_S, z_j)
924         R = phi_resid * z_j_resid
925         # GCM z-statistic
926         Tn = np.sqrt(len(R)) * R.mean() / R.std(ddof=1)
927         pval = 2.0 * scipy.stats.norm.sf(abs(Tn))
928         out.append((j, float(pval)))
929     return pd.DataFrame(out, columns=["candidate", "p_value"]).
930     set_index("candidate")

```

931 **Linear treatment-interaction test (baseline)** Fits Eq. (20) via OLS and tests the candidate's main  
932 effect and treatment interaction jointly.

```

933 import pandas as pd
934 import statsmodels.api as sm
935 def cci_test_linear(T, Y, Z, S, candidates):
936     df = Z[S].copy() if S else pd.DataFrame(index=Z.index)
937     df["_T"] = T.astype(int)
938     for s in S:

```

```

939     df[f"_T_x_{s}"] = df["_T"] * df[s]
940     out = []
941     for j in candidates:
942         df_j = df.copy()
943         df_j[j] = Z[j]
944         df_j[f"_T_x_{j}"] = df_j["_T"] * df_j[j]
945         model = sm.OLS(Y, sm.add_constant(df_j)).fit()
946         f_test = model.f_test([f"{j} = 0", f"_T_x_{j} = 0"])
947         out.append((j, float(f_test.pvalue)))
948     return pd.DataFrame(out, columns=["candidate", "p_value"]).
949         set_index("candidate")

```

## 950 **C Semi-synthetic experiments**

### 951 **D Application 1: LEAP1000 Programme (Ghana)**

### 952 **E Application 2: Youth Growth Program (Uganda)**